Experiment No : 4

Tutorial Title:. Implement Database Management using arrays of structures with operations Create, Display, Modify,Search and Sort.

Name of Student: Lohakare Shweta

Class: SY **Div:** 2 **PRN No:**B25ET1099 **Batch:**B2

Program:

```
#include <stdio.h>
struct employee
{  int no;
   char name[20];
   float salary;
};
int main()
{ struct employee e[50], temp;
   int n, i, j, ch, no, found;
   printf("Enter number of employees: ");
   scanf("%d", &n);
   for(i = 0; i < n; i++)
   { printf("Enter No Name Salary: ");
      scanf("%d %s %f", &e[i].no, e[i].name, &e[i].salary);  }
do
   {  printf("\n1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit");
      printf("\nEnter choice: ");
      scanf("%d", &ch);
    if(ch == 1)
      { for(i = 0; i < n; i++)
            printf("\n%d %s %.2f", e[i].no, e[i].name, e[i].salary);  }
        else if(ch == 2)
      { printf("Enter employee no: ");
         scanf("%d", &no);
         for(i = 0; i < n; i++)
            if(e[i].no == no)
            {   printf("Enter new name and salary: ");
               scanf("%s %f", e[i].name, &e[i].salary); }
             } else if(ch == 3)
      { printf("Enter No Name Salary: ");
```

```
        scanf("%d %s %f", &e[n].no, e[n].name, &e[n].salary);
        n++;
      } else if(ch == 4)
      {  printf("Enter employee no: ");
        scanf("%d", &no);
        found = 0;
for(i = 0; i < n; i++)
            if(e[i].no == no)
   { printf("%d %s %.2f", e[i].no, e[i].name, e[i].salary);
              found = 1; }
            if(found == 0)
            printf("Employee not found"); }
       else if(ch == 5)
      { for(i = 0; i < n - 1; i++)
            for(j = i + 1; j < n; j++)
              if(e[i].no > e[j].no)
              { temp = e[i];
                e[i] = e[j];
                e[j] = temp; }
               printf("Records sorted"); }
      } while(ch != 6);
return 0; }
```

Output:

main.c

Run ▶

```c
#include <stdio.h>
struct employee
{  int no;
   char name[20];
   float salary;
};
int main()
{ struct employee e[50], temp;
   int n, i, j, ch, no, found;
   printf("Enter number of employees: ");
   scanf("%d", &n);
   for(i = 0; i < n; i++)
   { printf("Enter No Name Salary: ");
      scanf("%d %s %f", &e[i].no, e[i].name, &e[i].salary);  }
do
   {  printf("\n1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit");
      printf("\nEnter choice: ");
      scanf("%d", &ch);
    if(ch == 1)
      { for(i = 0; i < n; i++)
            printf("\n%d %s %.2f", e[i].no, e[i].name, e[i].salary);  }
         else if(ch == 2)
      { printf("Enter employee no: ");
         scanf("%d", &no);
         for(i = 0; i < n; i++)
            if(e[i].no == no)
            {   printf("Enter new name and salary: ");
               scanf("%s %f", e[i].name, &e[i].salary); }
             } else if(ch == 3)
      { printf("Enter No Name Salary: ");
         scanf("%d %s %f", &e[n].no, e[n].name, &e[n].salary);
         n++;
      } else if(ch == 4)
      {  printf("Enter employee no: ");
         scanf("%d", &no);
         found = 0;
for(i = 0; i < n; i++)
            if(e[i].no == no)
```

Output

```
Enter number of employees: 2
Enter No Name Salary: 1 siya 1000
Enter No Name Salary: 2 ram 1200

1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice: 2
Enter employee no: 1
Enter new name and salary: Siyaa 10000

1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice: 2
Enter employee no: Ram 12000
Enter new name and salary:
1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice:
```

main.c

Run ▶

```c
do
   {  printf("\n1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit");
      printf("\nEnter choice: ");
      scanf("%d", &ch);
    if(ch == 1)
      { for(i = 0; i < n; i++)
            printf("\n%d %s %.2f", e[i].no, e[i].name, e[i].salary);  }
         else if(ch == 2)
      { printf("Enter employee no: ");
         scanf("%d", &no);
         for(i = 0; i < n; i++)
            if(e[i].no == no)
            {   printf("Enter new name and salary: ");
               scanf("%s %f", e[i].name, &e[i].salary); }
             } else if(ch == 3)
      { printf("Enter No Name Salary: ");
         scanf("%d %s %f", &e[n].no, e[n].name, &e[n].salary);
         n++;
      } else if(ch == 4)
      {  printf("Enter employee no: ");
         scanf("%d", &no);
         found = 0;
for(i = 0; i < n; i++)
            if(e[i].no == no)
   { printf("%d %s %.2f", e[i].no, e[i].name, e[i].salary);
               found = 1; }
            if(found == 0)
            printf("Employee not found"); }
      else if(ch == 5)
      { for(i = 0; i < n - 1; i++)
            for(j = i + 1; j < n; j++)
               if(e[i].no > e[j].no)
               { temp = e[i];
                  e[i] = e[j];
                  e[j] = temp; }
               printf("Records sorted"); }
      } while(ch != 6);
return 0; }
```

Output

```
Enter number of employees: 2
Enter No Name Salary: 1 siya 1000
Enter No Name Salary: 2 ram 1200

1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice: 2
Enter employee no: 1
Enter new name and salary: Siyaa 10000

1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice: 2
Enter employee no: Ram 12000
Enter new name and salary:
1.Display 2.Modify 3.Append 4.Search 5.Sort 6.Exit
Enter choice:
```